Lesson Notes

MODULE 7 | LESSON 2

PREDICTING DISASTER WITH PYTHON

Reading Time	30 minutes
Prior Knowledge	Inference, Bayesian networks
Keywords	d-separation, Markov blanket, Hard evidence, Soft evidence, Virtual evidence, Negative evidence, Jeffrey's rule

Wu et al. discuss their clever analysis at length in the last lesson's reading, but in this lesson, we walk through a notebook based on that analysis that accomplishes many of the same things.

1. Separation and Blankets

In this lesson, we reviewed the repo for the construction of the Bayesian network discussed in the last lesson, putting this powerful flood risk prediction tool in our own toolbox for inspection and modification. Now that we have seen several examples of more complex Bayesian networks, it may be helpful to review some of the concepts that were mentioned in earlier readings in light of the additional context. Specifically, these concepts help better understand and describe the chains of causality underlying the network. When a network is too small, as in the case of some of the toy Bayesian networks we started with (e.g. the Monty Hall and the oil stock price examples), these concepts hardly have enough nodes to illustrate themselves.

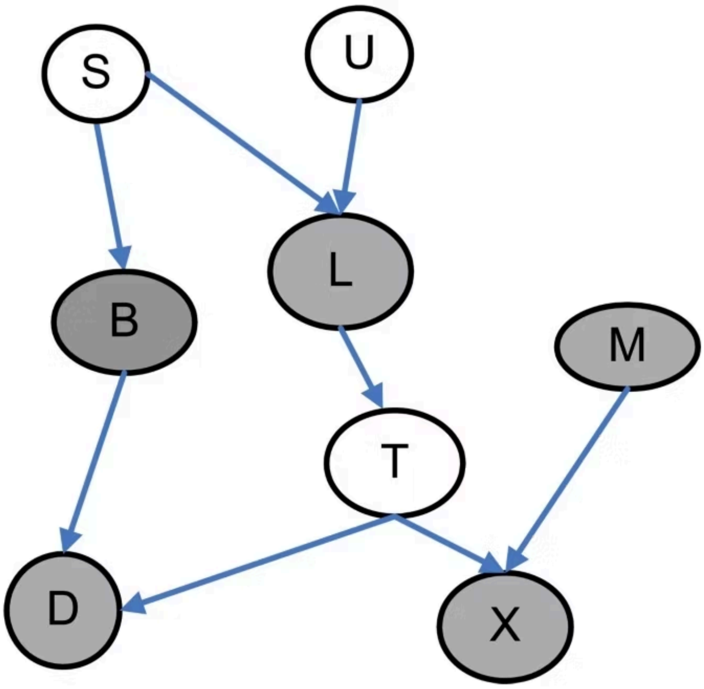
1.1 Markov Blankets

Markov blankets, for example, will be defined and discussed in more detail in the next module's reading, but let us note here that they consist of the minimal subset of nodes needed to make this the target node conditionally independent of all the other nodes in the model (BayesiaLab). Another way of saying this is that a target node's Markov blanket consists of all the "nodes that make the rest redundant when performing inference" on the target node (Scutari 91). In general, it consists of the following:

- The target node's parent nodes
- The target node's children nodes
- Any other nodes with the same children nodes

Note the significant node that does not form part of the Markov blanket: the target node itself.

Figure 1: Markov Blanket (for Node T)



Source: Han, Bing, et al. "A Markov Blanket-Based Method for Detecting Causal SNPs in GWAS." *BMC Bioinformatics*, vol. 11, no. 3, 2010, https://doi.org/10.1186/1471-2105-11-S3-S5.

1.2 D-Separation

By nature of the dependence/independence relationship, D-Separation is fundamentally related to the Markov blanket. In fact, the Markov blanket is just the set of nodes that "d-separate" the target node from the rest of the network (Carraro 31). Hopefully, this explains why we introduce the easier to understand Markov blanket before the d-separation concept itself. The "d" here, by the way, refers to "directed separation."

Again, we see the formal definition and in-depth description in the next module, but for now, let it suffice to say that it refers to the idea that "graphical separation implies probabilistic independence" in a Bayesian network (Scutari 21). One can also say:

"d-separation is a criterion for deciding, from a given a causal graph, whether a set X of variables is independent of another set Y, given a third set Z. The idea is to associate 'dependence' with 'connectedness' (i.e., the existence of a connecting path) and 'independence' with 'unconnected-ness' or 'separation'" (Pearl).

2. Likelihood Weighting with Soft, Virtual and Negative Evidence

Spolaor et al. make clear in the comments on their code that the pgmpy library was not sufficient for the weighted sampling approach they wanted to take, resulting in the need to extend two of that library's classes with the following:

- class ExtendedApproxInference(ApproxInference), which extends pgmpy's ApproxInference; and
- class ExtendedBayesianNetwork(BayesianNetwork), which extends pgmpy's class for Bayesian Models, allowing the simulation of data from the model with Weighted Likelihood.

They note that this latter extended class uses methods from `pgmpy.sampling.Sampling.likelihood_weighted_sample` to generate the data, but the function `simulate_by_weighted_likelihood` clearly takes a few more arguments compared to pgmpy's `likelihood_weighted_sample`

```
n_samples=10,

        do=None,

        evidence=None,

        virtual_evidence=None,

        virtual_intervention=None,
```

```
include_latents=False,

partial_samples=None,

seed=None,

show_progress=True,
```

where the bold underlined arguments are new in the notebook's extended class. As you can see, much of the change is related to the ideas of "virtual evidence" and "virtual intervention." Surprisingly, although Spolaor et al. went through the trouble of extending the two classes to allow for these arguments, they do not seem to exploit the addtional functionality in the notebook itself. What's more, both of these functionalities appear available in the pgmpy documentation. See them in action here: https://pgmpy.org/examples/Simulating%20Data.html#Simulation-with-soft/virtual-evidence.

The comments by Spolaor et al. raise another issue. The repeated use of "virtual/soft" to describe evidence and intervention seem to suggest that "virtual" and "soft" can be used interchangeably in this context. In fact, more discerning authors carefully distinguish between hard, soft, and virtual evidence; some also add additional categories of fuzzy evidence to this mix. We will not broach fuzzy evidence types, which invoke Fuzzy Logic Theory, but let us take the opportunity to get clear about the distinctions among these other three types of evidence.

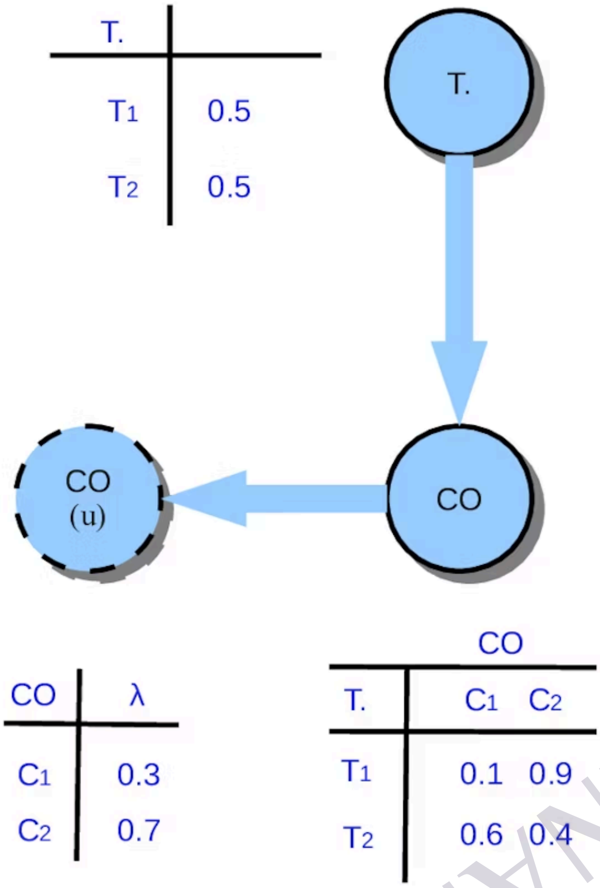
Actually, we can join Sturlaugson and Sheppard by starting with even broader categorizations of evidence: 1.) certain positive evidence, 2.) uncertain positive evidence, and 3.) certain negative evidence. In this case, certain positive evidence refers to traditional *hard evidence*, where "the observations are trusted with complete confidence" and inference merely requires the straightforward calculation of posterior probabilities (Sturlaugson and Sheppard 103). Crucial for this evidence category is that once the state of a variable is observed, the probability associated with that state of the variable is one; this conception opens the door to the more generalized version, uncertain evidence, in which the probabilities of observed states can be less than one (Sturlaugson and Sheppard 103).

Within uncertain evidence, we find both virtual and soft evidence. Mrad et al. admit that the two are often confused or conflated, but distinguish them as follows:

- **Virtual evidence** is "evidence with uncertainty, and can be represented as a likelihood ratio" and for this reason "is also called likelihood evidence" (Mrad 40). The likelihood "is a real number in [0, 1]" (Mrad 43). Building on the point made at the end of the previous paragraph, hard evidence would have a likelihood of one.
- **Soft evidence**, on the other hand, is "evidence of uncertainty" and "is represented as a probability distribution of one or more variables" (Mrad 40, 44). With just a few transformations, soft evidence can be converted into virtual evidence (Mrad 44).

In practice, virtual evidence is established by assigning a new child node to the observed node, where the conditional probabilities of this new child node match the strength of the evidence—i.e., the likelihoods—and the ratio of likelihoods represent the confidence that the actual observation indeed corresponds to the observed state. In this way, virtual evidence "weights the marginal probabilities by the likelihood of the evidence" (*Sturlaugson* and Sheppard 103), which is what we want.

Figure 2: Example of Virtual Evidence



Source: Xiang, Yun, et al. "Mobile Sensor Network Noise Reduction and Recalibration using a Bayesian Network." *Atmospheric Measurement Techniques*, vol. 9, 2016, pp. 347–357, https://www.proquest.com/openview/bb67ea88d37cfafeb8e8ee92b63ce128/1?pq-origsite=gscholar&cbl=105742.

Soft evidence, on the other hand, invokes **Jeffrey's rule** which, conditioned on some event, revises "a probability measure by another probability function" (Smets 500). For our present purposes, we shall set aside the requisite considerations of the probability space with its Boolean algebra implied by reliance on Jeffrey's rule; instead, we will note that soft evidence simply transforms the marginal distribution of X from P(X) to P'(X) such that it conforms exactly to the probability of the evidence, while virtual evidence does not require any such distributional update (Sturlaugson and Sheppard 103). Naturally, since the two approaches specify uncertain evidence in such different ways, they generally yield different probabilities, which is all the more reason not to confuse them.

3. Negative Evidence

For completeness' sake, in keeping with the Sturlaugson's categorization, we mention negative evidence as "a special case of uncertain positive evidence" (103). That is, if we do not observe the variable in a certain state, this counts as uncertain evidence that there is zero probability of that state, in which case the state could be removed from the probability space and the remaining probabilities would then be re-normalized (Sturlaugson and Sheppard 104).

While the use of virtual, soft, and indeed negative evidence in Bayesian network modeling is advanced—even, Spolaor et al. do not appear to make use of it—and aside from the fact that the concepts are interesting in and of themselves, it is noteworthy that Bayesian networks can be designed to incorporate these epistemological and probabilistic subtleties.

4. Conclusion

In this lesson, we walk through a Python notebook that illustrates the Bayesian network modeling and predictive analysis that we read about it in the previous lesson. In the next two lessons, we set aside Bayesian networks temporarily to focus on the incorporation of climate risk in credit models, in line with the reading by Wouters in the last lesson.

References

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